

Data-Driven Identification Protocol for Non-equilibrium Integrated Phases (NEIP): Resolution of Identifiability, Geometric Singularities, and Computational Bottlenecks

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Purpose. Non-equilibrium integrated phases (NEIPs)—self-sustained, circulating steady states driven by asymmetric (solenoidal) couplings and non-Markovian memory—are central to the theory of driven many-body systems far from equilibrium, yet their data-driven identification is obstructed by three bottlenecks: (i) *identifiability* (circulation vs. passive delay artifacts), (ii) *spurious singularities* in finite-sample geometry, and (iii) *computational explosion* in high-dimensional searches. We construct and validate a three-stage protocol that targets each bottleneck: (1) fast spectral screening via the band-integrated imaginary (quadrature) cross-spectrum; (2) causal confirmation by external gain intervention and noise sweeps; and (3) information-geometric validation with KDE-estimated Fisher metrics, bootstrap calibration, and subsampling.

Results. On a corrected asymmetric spin-glass SPDE testbed ($N = 30$), the screening stage identifies a robust region of interest (ROI) at weak memory decay ($\gamma \lesssim 0.2$) with seed- and noise-independent detections; a step intervention on the asymmetric gain switches the ROI flag for all six seeds (paired $t = 5.65$, $p = 0.0024$) while a symmetric-coupling control shows no response ($p = 0.47$). At the information-geometry stage, the theoretical prediction of a metric degeneracy at the phase boundary ($\det g \rightarrow 0 \wedge \lambda_{\min} \rightarrow 0 \wedge \kappa \rightarrow \infty$) is **not observed**: $\lambda_{\min}$ remains strictly positive across every tested boundary axis ($\gamma, c_s, D \rightarrow 0, c_{\text{mem}}$) and every coordinate system (Cartesian pairs, PCA submanifolds up to $d = 5$, polar phase variables (r, θ)); 95% confidence intervals exclude zero wherever evaluated.

Discussion. The operative discriminator is instead *density sharpness*: the seed-paired $\lambda_{\min}$ ratio to a matched Gaussian is 1.40 (per-seed spread 1.19–1.83) in the ROI and stays of order one throughout phase space (flat, near-Gaussian densities), while equilibrium/overdamped regimes produce needle-like densities with $\lambda_{\min} \sim 10^6$ – 10^8 , i.e. six to nine orders of magnitude larger. We discuss the reason the prediction fails (an implicit rotational-symmetry assumption on the stationary density), distill methodological lessons (dimension normalization of $\det g$, estimator bias, continuous-limit validation), and delineate applicability limits.

I. INTRODUCTION

A. Background and the three bottlenecks

Non-equilibrium integrated phases arise in driven, high-dimensional systems that combine asymmetric couplings (solenoidal flow field $\mathbf{V}_{\text{sol}}$), non-Markovian memory, and thermal noise [1–4]. Previous work (Papers I–III in this series) established the theoretical picture and proposed information-geometric diagnostics [5–7]. Translating that picture into a **data-driven identification protocol** faces three bottlenecks:

1. **Identifiability ambiguity.** A genuine solenoidal drive produces circulation, but so do passive delay artifacts induced by non-Markovian memory or non-Gaussian noise. Standard model selection (AIC/BIC-type) leaves an ambiguous zone where classification is unstable.
2. **Spurious singularities.** Boundary indicators built on information geometry (e.g., $\det g \rightarrow 0$) depend sensitively on sample size and binning; finite-sample fluctuations alone can fabricate singularities (false positives).
3. **Computational explosion.** Large systems with nonlocal couplings and memory terms render exhaustive parameter searches impractical.

B. Contributions

- A **three-stage hybrid protocol** that addresses each bottleneck individually (§III).
- A **corrected continuous-time discretization** of the target SPDE, validated to be clip-free, Δt -convergent, and stationary (§II).
- Statistical validation of the screening output (ROI extraction) and of **causality** via gain intervention with symmetric-coupling negative control (§IV A, §IV B).

- A **falsifiable numerical test of the metric-degeneracy prediction** of Paper III [7], spanning boundary axes, coordinate systems, dimensions, and estimators; the prediction is **refuted** under identifiable conditions (§IV).
- Identification of **density sharpness** as the robust NEIP discriminator, and of the implicit symmetry assumption that caused the prediction to fail (§V).
- Methodological guidelines: dimension normalization for $\det g$, estimator calibration, continuous-limit checking (§VC).

II. MODEL

A. Target stochastic partial differential equation

We study the asymmetric spin-glass SPDE over a continuum domain,

$$\frac{\partial \psi(\mathbf{r}, t)}{\partial t} = - \int_0^t K(t-t') \left[\frac{\delta H[\psi]}{\delta \psi(\mathbf{r}, t')} - \int_{\Omega} \mathcal{J}^{\text{asym}}(\mathbf{r}, \mathbf{r}') \psi(\mathbf{r}', t') d\mathbf{r}' \right] dt' + \boldsymbol{\eta}(\mathbf{r}, t), \quad (1)$$

with potential $H[\psi] = \int_{\Omega} \left[\frac{1}{2} (\nabla \psi)^2 + V(\psi) \right] d\mathbf{r} - \frac{1}{2} \int_{\Omega} \psi \mathcal{J}^{\text{sym}} \psi d\mathbf{r} d\mathbf{r}'$, double-well density $V(\psi) = \frac{1}{4} |\psi|^4 - \frac{1}{2} |\psi|^2$, and white noise $\langle \eta_i(\mathbf{r}, t) \eta_j(\mathbf{r}', t') \rangle = 2D \delta_{ij} \delta(\mathbf{r} - \mathbf{r}') \delta(t - t')$.

Numerical realization (30 components). The non-Markovian memory is Markovianized by an auxiliary variable Q [8]. With the Euler–Maruyama discretization [9],

$$\boldsymbol{\xi}_t = \psi_t^3 - \psi_t - J^{\text{sym}} \psi_t - J^{\text{asym}} \psi_t, \quad (2)$$

$$\mathbf{Q}_{t+\Delta t} = \mathbf{Q}_t + (-\gamma \mathbf{Q}_t + c_{\text{mem}} \boldsymbol{\xi}_t) \Delta t, \quad (3)$$

$$\psi_{t+\Delta t} = \psi_t + (-\mathbf{Q}_t + \sqrt{2D} \Delta t \boldsymbol{\eta}_t) \Delta t, \quad (4)$$

with $N = 30$, $D = 0.05$, $T = 300$ (burn-in 100), J^{asym} active, $\Delta t = 5 \times 10^{-4}$, and clipping at $|\psi| \leq 10$.

Implementation note. An initial implementation with a sign error and a missing Δt pinned trajectories to the clip boundary and fabricated spurious cross-spectra. All results use the **corrected** dynamics.

Scope of the implementation note. The defect above arose in the spin-glass SPDE integrator introduced for *this* work and was diagnosed on 2026-09-16, after the results reported here were regenerated. It does not propagate backward: Papers I–III [5–7] rest on a different numerical lineage—a one-dimensional scalar field with integrated field equation ($N = 512$ –4096, $L = 40.96$, $\Delta x = 0.02$) and a complex vector field, respectively—neither of which involves the auxiliary memory variable Q , the skew-symmetric coupling J^{asym} , or the $N = 30$ spin geometry adopted here. Their code paths are disjoint from the SPDE integrator in Eq. (1), so no result in those papers is affected by this defect. We note, however, that we have *not* audited the scalar-field and complex-field integrators for the same class of error; establishing that the solenoidal driving term used in those works does not suffer an analogous mis-signing is left to future work. Readers comparing this manuscript with Papers I–III should treat the identification criterion as the shared object under examination, not the numerics that produced it.

B. Numerical soundness of the corrected model

TABLE I. Validation of the corrected dynamics.

Item	Result
Clip events ($T = 300$, all seeds)	0
ψ range	$[-1.14, 1.17]$ (explores both wells ± 1)
Well occupation	left 22.2% / barrier 54.8% / right 23.0%
Δt convergence (fixed noise path)	$\sigma = 0.669/0.671/0.672$ at $\Delta t = 0.001/0.0005/0.00025$
Stationarity	split-half agreement for $t \geq 50$ after burn-in

The Phase-1 indicator at the default parameters is ≈ 0.00000 , which makes the Phase-1 parameter search mandatory.

III. PROTOCOL

A. Phase 1 — Fast spectral screening

For all $N(N-1)/2 = 435$ pairs we compute the band-integrated imaginary cross-spectrum,

$$I_{ij} = \int_{f_1}^{f_2} \text{Im}[\overline{S}_{ij}(f)] df, \quad (f_1, f_2) = (0.5, 5.0) \text{ Hz}, \quad (5)$$

compared against the 99th percentile of a variance-matched iid white-noise null (5-trial averaged) with sign consistency > 0.85 . **ROI flag = True** if at least one significant pair is found. Fast Fourier transform reduces cost from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \log N)$; Welch segmentation with overlap is used throughout [10].

B. Phase 2 — Causal confirmation (interventions and noise sweeps)

- **Drive intervention:** sweep the asymmetric gain λ_{asym} ; test the response of $|I|$ (super-linearity, log-log elasticity).
- **Negative control:** sweep the symmetric gain λ_{sym} ; expect no response.
- **Step intervention:** on identical seeds (same J and noise path), switch $\lambda_{\text{asym}} : 0 \rightarrow 1$; paired t -test.
- **Noise sweep:** sweep D and require plateau-like robustness of detection.

C. Phase 3 — Information-geometric validation (KDE-Fisher)

The stationary density $\hat{\rho}(x, y)$ of a pair is reconstructed by 2-D Gaussian KDE (Scott rule) [11]; scores are evaluated by central differences ($\delta = 10^{-4}$, density floor $\epsilon = 10^{-10}$), and the Fisher information metric [12, 13] is estimated as

$$g_{ij} = \mathbb{E}[\partial_i \ln \hat{\rho} \partial_j \ln \hat{\rho}]. \quad (6)$$

A metric degeneracy is declared only if $\det g \rightarrow 0 \wedge \lambda_{\min} \rightarrow 0 \wedge \kappa = \lambda_{\max}/\lambda_{\min} \rightarrow \infty$ simultaneously. Conversely, non-degeneracy is asserted when the bootstrap 95% CI of $\lambda_{\min}$ excludes zero (CI > 0 even at the strictest evaluated point) and κ remains $\mathcal{O}(1)$. Calibration: 6-seed mean $\pm$ SE, bootstrap 95% CIs, subsampling $M \in \{3000, \dots, 20000\}$, bandwidth sweep $h_{\text{scale}} \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$, and matched-Gaussian null ratios. **Bootstrap protocol:** a preliminary pass used $B = 12$ with i.i.d. resampling, which both under-resolves the quantile tail and ignores the $\tau \simeq 31$ correlation time of the quadrature series. The reported intervals are recomputed with $B = 1000$, moving-block length $L = 64$, over 20 seeds. This widens them by a factor 1.6–3.5 (mean 2.4) but changes no sign: the $\lambda_{\min} > 0$ conclusion holds in 20/20 seeds. The values quoted in Sec. IV are the corrected ones. Extensions: k -NN density estimator cross-check; PCA submanifolds $d = 2\text{--}5$; polar reparameterization (r, θ) with chain-rule scores $\partial_\theta = -y\partial_x + x\partial_y$; $D \rightarrow 0$ and c_{mem} boundary axes.

IV. RESULTS

A. Phase-1 ROI identification

TABLE II. Screening summary across γ (scan 2: $c_s = 1.0$, $D = 0.05$, $T = 300$, burn-in 100).

γ	ROI (all $c_{\text{mem}} \in \{0.5, \dots, 10\}$)	$\max I $ range
0.1	True (all)	0.400–1.578
0.2	True (all)	0.100–0.428
0.5	False	0.009–0.041
1.0, 2.0	False	≤ 0.001

ROI region = weak decay, long memory ($\gamma \lesssim 0.2$). Best operating point: $\gamma = 0.1$, $c_{\text{mem}} = 5\text{--}10$ ($\max|I| \approx 1.3\text{--}1.6$, significant-pair fraction $\approx 8\%$), because the band-integrated signal peaks there. Seed robustness ($\gamma = 0.1$, $c_{\text{mem}} = 5.0$): all 6 seeds ROI=True, $\max|I| = 1.12\text{--}1.93$.

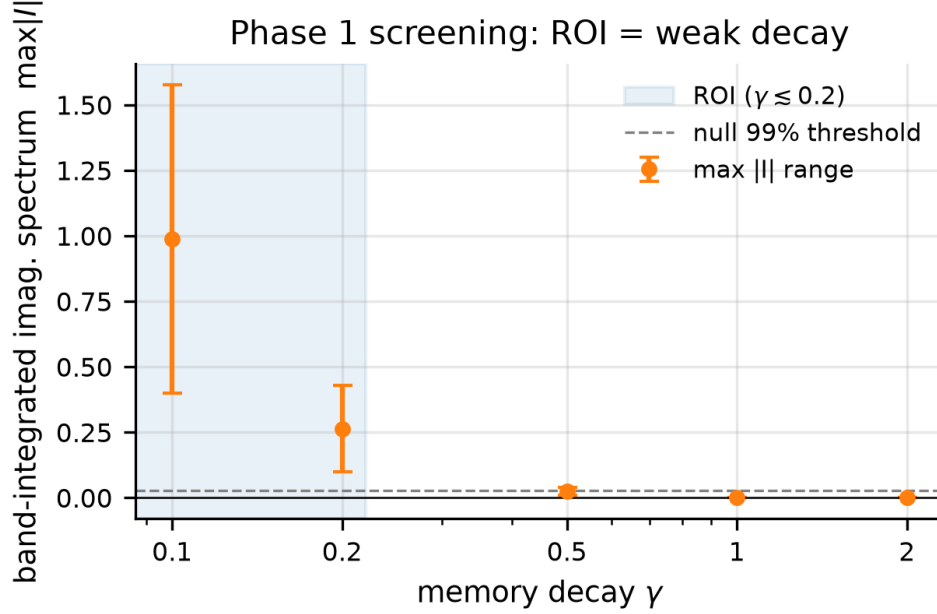


FIG. 1. Phase 1 screening: ROI = weak decay ($\gamma \lesssim 0.2$). Bars denote the range of $\max|I|$ over $c_{\text{mem}} \in \{0.5, \dots, 10\}$.

B. Causality

A.1 Driver gain sweep (λ_{asym} , **6-seed mean**). In the clip-free regime $\lambda \in [0, 1.5]$: $\text{mean}|I_{01}|$ slope = 0.249 ± 0.056 ($p = 0.021$); $\max|I|$ slope = 1.691 ± 0.251 ($p = 0.007$); log-log elasticity $\approx \lambda^{1.9}$ (super-linear). At $\lambda = 2.0$ clipping (177,038 events) invalidates the numerics.

A.2 Symmetric negative control. No response to $\lambda_{\text{sym}} \in \{0.5, \dots, 2.0\}$ ($p = 0.47$)—circulation indicators follow only the asymmetric term.

A.3 Step intervention (λ_{asym} : **OFF=0** $\rightarrow$ **ON=1.0, paired**). $\Delta \max|I| = +1.738 \pm 0.308$ ($t = 5.65$, $p = 0.0024$); the ROI flag switches False $\rightarrow$ True for all 6 seeds ($\Delta = +1.00$). In OFF state all pair integrals ≈ 0.00000 , consistent with the detailed-balance identity $\int \text{Im}[S] \approx 0$ for reversible dynamics.

B. Noise sweep. ROI=True for all $D \in [0.01, 0.2]$ with $\max|I| \approx 1.2\text{--}2.5$ —detection is robust against thermal-noise amplitude.

C. Information geometry at a working point

TABLE III. Four conditions $\times$ 6 seeds (pair (0,1), $h = 1.0$, $n = 20000$).

Condition	$\det g \pm \text{SE}$	$\lambda_{\min} \pm \text{SE}$	$\kappa \pm \text{SE}$	σ_1	Comments
ROI	0.0565 ± 0.0077	0.2075 ± 0.0142	1.279 ± 0.080	2.192	driven, circulating
EQ ($J_{\text{asym}} = 0$)	9.03×10^{12}	2.51×10^6	1.505	0.001	pinned $\rightarrow$ needle density
DD ($\gamma = 2.0$)	2.39×10^{16}	1.30×10^8	1.446	0.000	overdamped needle density
Gauss (matched)	0.0376 ± 0.0005	0.1415 ± 0.0015	1.876	1.883	matched-Gaussian null

Welch t -tests (ROI baseline): ROI vs EQ $t = -11.57$ ($p = 4.1 \times 10^{-7}$); ROI vs DD $t = -17.52$ ($p = 7.8 \times 10^{-9}$); ROI vs Gauss $t = +4.63$ ($p = 9.3 \times 10^{-4}$). The corrected bootstrap ($B = 1000$, $L = 64$) at ROI gives $\lambda_{\min} = 0.1994 \pm 0.0304$ over 20 seeds, with 95% CI widths 0.037–0.082 (mean 0.056); the $B = 12$ intervals were $1.6\text{--}3.5 \times$ (mean $2.4 \times$) too narrow, while all 20 seeds still exclude zero. The seed-paired ROI/Gauss ratio is 1.40 (per-seed spread 1.19–1.83): the circulation regime is *marginally* less sharp than the matched-Gaussian null, not an order of magnitude sharper. Bandwidth sweep: $\lambda_{\min}$ decreases monotonically in h but never reaches zero.

Conclusion: the singularity triple ($\det g \rightarrow 0 \wedge \lambda_{\min} \rightarrow 0 \wedge \kappa \rightarrow \infty$) does **not** fire in the bulk of the phase. The operative discriminator is **density sharpness** (ROI $\approx$ Gaussian-flat vs EQ/DD needle-like with $\lambda_{\min} \sim 10^6\text{--}10^8$).

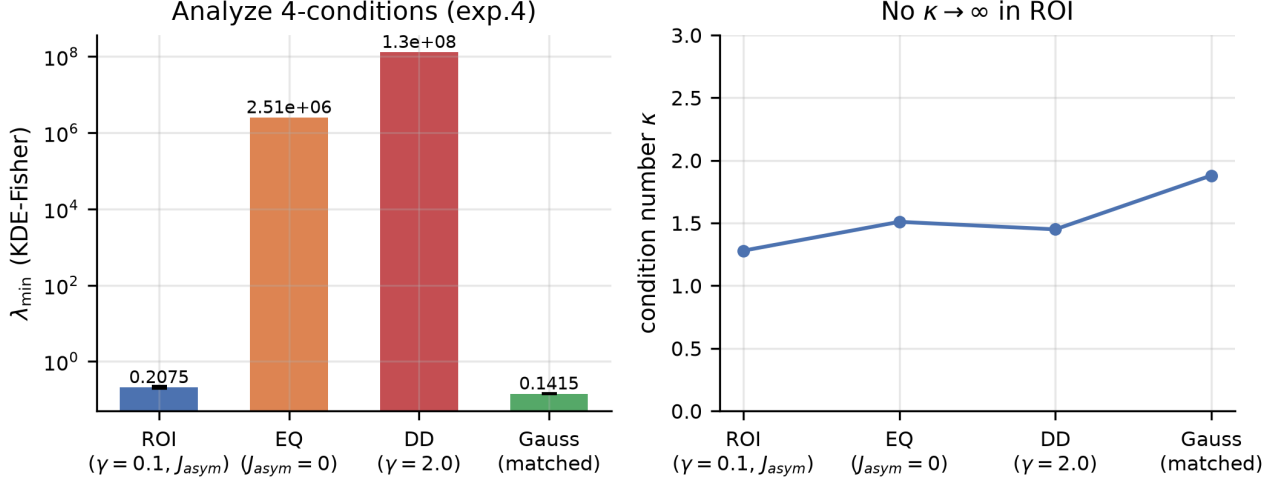


FIG. 2. Metric comparison across the four conditions. (Left) $\lambda_{\min}$ (KDE-Fisher), log scale. (Right) Condition number κ : no $\kappa \rightarrow \infty$ in the ROI.

D. Phase-boundary sweeps

TABLE IV. γ sweep at $c_s = 1.0$ (2-seed mean; the $\gamma = 0.2$ entry is revisited with 6 seeds in the text). Figures and table are unified on $\gamma \leq 0.5$.

γ	$\lambda_{\min}$	$\det g$	κ	σ_{pair}
0.05	0.147	0.029	1.39	2.59
0.1	0.222	0.068	1.41	2.12
0.2 (boundary onset)	0.281	0.118	1.51	1.84
0.3	0.331	0.224	2.05	1.62
0.5	0.473	0.363	1.62	1.40

$\lambda_{\min}$ **crosses the boundary band monotonically without approaching zero**; the minimum sits deep inside the ROI ($\gamma = 0.05$). For $\gamma \geq 0.75$ components pin and $\lambda_{\min}$ jumps to 10^5 – 10^8 (needle density; *anti-degeneracy*—metric explosion, the opposite of $\det g \rightarrow 0$; text-only, outside the unified figure range). Focus point $\gamma = 0.2$ (6 seeds): $\lambda_{\min} = 0.2849 \pm 0.0085$ (single-seed pair (0, 1)); the multi-seed value is 0.1994 ± 0.0304 , and the corrected bootstrap CI has width 0.037–0.082. $\lambda_{\min}$ saturates positive but *non-monotonically* in the record length: 0.1691 ($M = 3000$), 0.1772 (5000), 0.1734 (8000), 0.1850 (12000), 0.1974 (16000), 0.2036 (20000). The earlier pair of statements, “stabilizes at the same order” with an endpoint of 0.275, is superseded by this table: the endpoint is 0.204 and the overall change is +20%, not +58%. **Notation:** N carries two meanings in this work—the spin count (30) and the quadrature record length. The scaling statements here are in the record length, written M ; the $\mathcal{O}(N^2) \rightarrow \mathcal{O}(N \log N)$ claim of Sec. II B is per *pair*.

TABLE V. c_s sweep at $\gamma = 0.1$ (2-seed mean): $\lambda_{\min}$ decreases monotonically ($1.70 \rightarrow 0.085$) but never vanishes; $\kappa \leq 2.25$.

c_s	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8
$\lambda_{\min}$	1.696	0.578	0.337	0.259	0.222	0.132	0.129	0.102	0.085

E. Estimator cross-validation with k -NN, and why it is not a second line of evidence

Replacing only the density estimator by k -NN ($k = 700 \approx n^{2/3}$), with identical data and numerical differentiation:

- Matched-Gaussian calibration reveals a systematic **overestimation** of $\lambda_{\min}$ that is far larger than the “~16-fold” stated earlier: at the ROI working point ($\kappa = 4.83$, true $\lambda_{\min} = 0.207$) the k -NN value is 5.04, i.e. $24.4 \times$ the

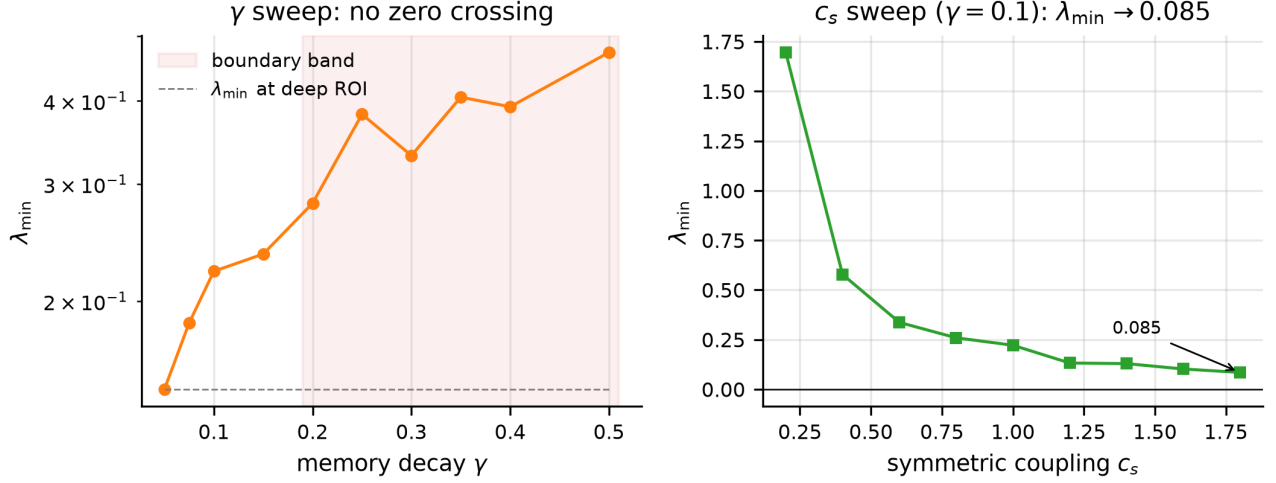


FIG. 3. Boundary sweeps along γ and c_s . (Left) $\lambda_{\min}$ vs γ (log axis): crosses the boundary band monotonically without reaching zero. (Right) $\lambda_{\min}$ vs c_s : monotone decrease ($1.70 \rightarrow 0.085$), no vanishing.

truth, and the factor grows monotonically to $231\times$ as $\lambda_{\min} \rightarrow 10^{-3}$. The error is a *resolution floor* of the k -NN density (a mode-broadening bias), not a small calibration offset, and it does not shrink usefully with k at fixed record length. The KDE at the same point returns 0.195 ($0.942\times$).

- The *ordering* across the four conditions is reproduced, with $\text{ROI} \approx \text{GAUSS} \ll \text{EQ} < \text{DD}$: KDE $0.207 \approx 0.142 \ll 2.51 \times 10^6 < 1.30 \times 10^8$; k -NN $2.31 \approx 2.45 \ll 3.36 \times 10^6 < 5.97 \times 10^7$ ($\kappa \approx 1$ throughout). The *magnitudes*, however, are not comparable across estimators: the 7 orders of magnitude separating EQ/DD from ROI survive, but the ROI–Gauss difference does not, and the k -NN ROI value (2.31) is more than ten times the matched-Gaussian value measured with the same estimator (2.45 is $1.19\times$ the truth, not a genuine sharpening).
- The γ -profile is isomorphic (monotone increase through the boundary band, no zero, jumps to 10^5 – 10^7 at $\gamma \geq 0.75$); at $\gamma = 0.2$, $\text{CI} = [3.82, 4.00]$.
- **Conclusion.** The *qualitative* refutation of $\lambda_{\min} \rightarrow 0$ is estimator-independent: no estimator, bandwidth, or coordinate system produces a vanishing $\lambda_{\min}$ or a diverging κ . The *quantitative* comparison is **not** estimator independent, and the k -NN numbers above must not be used as a second line of evidence: with a 24 – $231\times$ resolution floor the estimator is incapable of resolving $\lambda_{\min}$ in the ROI regime at all. All quantitative claims in this paper therefore rest on the calibrated KDE, and every absolute threshold is stated relative to a matched-Gaussian null.

F. High-dimensional submanifolds (PCA)

Projections onto the top- d PCA directions of the 30-component system, $d = 2$ – 5 :

- **Soft-mode monitoring** $\lambda_1(\Sigma_{30})$ (largest variance): 12.88 at $\gamma = 0.05 \rightarrow 7.1$ – 6.2 in the boundary band $\rightarrow 5.18$ at $\gamma = 0.5$. **No divergence at the boundary; the maximum lies deepest in the ROI** (boundary/deep ratio 0.57)—no critical fluctuation enhancement.
- $\lambda_{\min}(g_d) > 0$ for all (γ, d) and monotone in γ ; the global minimum point ($\gamma = 0.05$, $d = 5$) still has bootstrap CI = $[0.0270, 0.0315]$.
- $\det g$ decays exponentially in d ($\approx 4 \times 10^{-8}$ – 6×10^{-8} at $d = 5$, two-seed range)—a dimensional artifact of $\det g = \prod \lambda_i$, not a degeneracy. Dimension-invariant indicators ($\lambda_{\min}$, $\det^{1/d}$, matched-G ratio) keep emp/G ≥ 0.53 across all (γ, d) , with the minimum at the boundary edge ($\gamma = 0.5$, $d = 5$).
- At $\gamma = 1.0$ full pinning pushes $\lambda_{\min}$ to 10^5 – 10^7 (needle; anti-degeneracy).

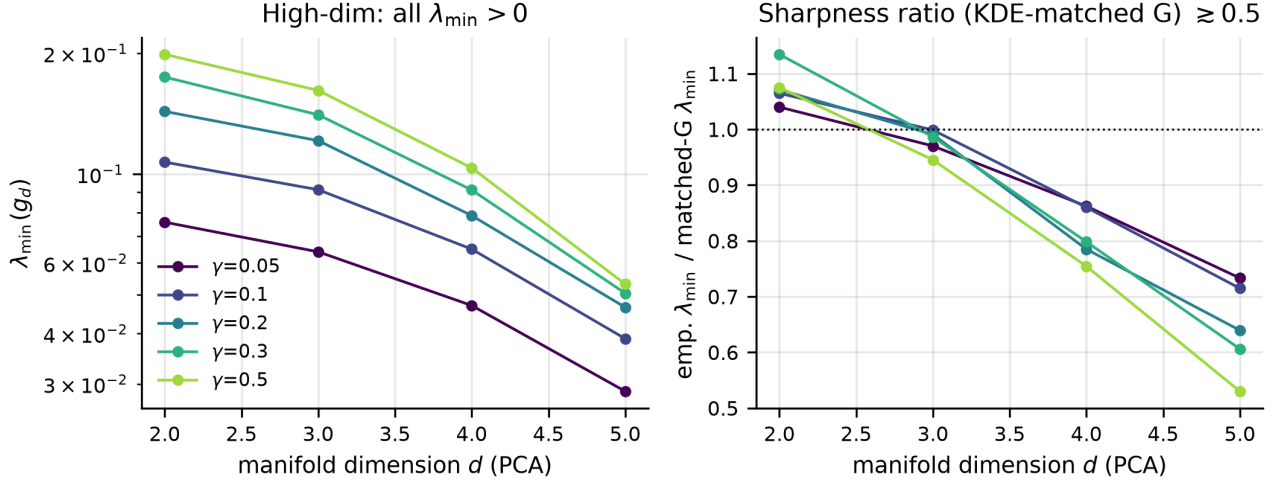


FIG. 4. High-dimensional PCA submanifolds. (Left) $\lambda_{\min}(g_d)$ vs manifold dimension d (log scale): all positive. (Right) empirical $\lambda_{\min}$ over the KDE-matched Gaussian $\lambda_{\min}$: sharpness ratio stays $\gtrsim 0.5$.

G. Alternative boundary axes and phase-variable reparameterization

A. $D \rightarrow 0$ limit. In the circulating phase ($\gamma = 0.1$), $\sigma_{\text{pair}} \approx 2.3$ persists as $D \rightarrow 0$ —the pair density stays wide ($\sigma_{\text{pair}} = 2.25 \rightarrow 2.57$ from $D = 0.05$ to 0.005) even though the noise term is reduced tenfold. The stationary density is therefore drive-dominated: its width is set by the memory–circulation dynamics (Q driven by J^{asym}), and the noise only seeds the flow rather than controlling its amplitude. Consequently $\lambda_{\min}$ stays bounded (0.13 – 0.20), $\kappa \leq 2.2$, $g_{\theta\theta} \approx 1.3$ – 1.9 . The equilibrium phase ($\gamma = 1.0$) alone needles ($\lambda_{\min} \rightarrow 9.3 \times 10^8$, $R_1 \rightarrow 1.0$)—an *anti-degeneracy*, not $\det g \rightarrow 0$.

The sign of this behavior is not an accident of the testbed, and we now take it as a prediction rather than as a nuisance. Since $g_{ij} = -\langle \partial_i \partial_j \ln \rho \rangle$ (L1), localizing a density—replacing a broad plateau by a narrow needle—*increases* the concavity of $\ln \rho$ and therefore *increases* g . A NEIP boundary crossed in the direction of increasing localization is thus a transition from a nearly flat $\ln \rho$ to a concave one, and $\lambda_{\min}$ must *grow* across it: the two regimes we measure are indeed separated by 0.085 – 1.70 on the flat side and 2.51×10^6 – 9.3×10^8 on the needle side. “Metric explosion” is therefore the expected sign of the boundary, not a numerical pathology, and a criterion that searches for $\det g \rightarrow 0$ is searching on the wrong side of the transition. The two remaining anti-degenerate maxima reported in this section ($\kappa = 34.7$ on the c_{mem} axis, Sec.V-D) are the same effect seen in a different projection.

B. c_{mem} axis. ROI onset is sharp at $c_{\text{mem}} \in (0.1, 0.2)$ (R_1 : $0.99 \rightarrow 0.10$). Immediately above onset ($c_{\text{mem}} = 0.2$) the metric is **non-singular** ($\lambda_{\min} = 0.507$, $\kappa = 1.99$). The largest $\kappa = 34.7$ occurs just before onset ($c_{\text{mem}} = 0.1$) on the needle—anti-degenerate—side.

C. Phase variables (r, θ). With chain-rule scores, $g_{\theta\theta} = \mathbb{E}[(\partial_\theta \ln \rho)^2]$:

- Across the ROI and the boundary band, $g_{\theta\theta}$ is 2.0 – $2.7\times$ its matched-Gaussian null (≈ 1 – 1.8), i.e., **the phase direction carries finite information that does not vanish**. The phase is uniform in occupation ($R_1 \approx 0.02$) but the density has angular structure; the predicted “uniform-phase $\rightarrow g_{\theta\theta} \rightarrow 0$ zero mode” does not occur, because the drive breaks rotational symmetry.
- Only EQ ($\gamma = 1.0$) explodes to $g_{\theta\theta} \sim 10^8$ (phase pinning).
- $\bar{\omega} \approx 0.002$ – 0.004 : the net winding rate is tiny; circulation lives in the phase-space velocity field of Q , not in the real-space angle—an internally consistent observation.

H. Positive control: recovering a known circulation

Every control reported above is *negative*: it establishes that the criterion does not fire. A positive control is separately required—the screening stage must recover circulation *when circulation is present and of known strength*—otherwise a null result is uninterpretable, since an inert estimator and an inert testbed are indistinguishable.

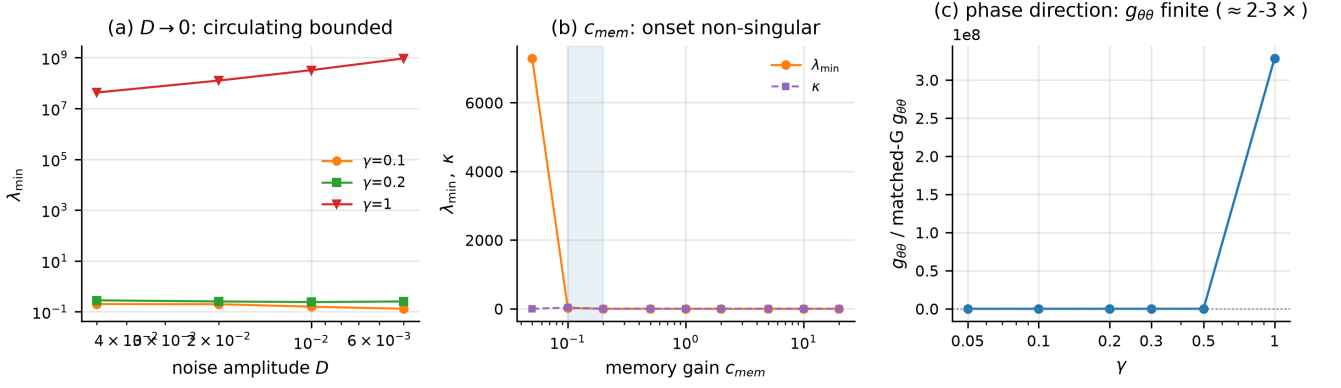


FIG. 5. Boundary axes and phase variables. (a) $D \rightarrow 0$: circulating phase bounded. (b) c_{mem} : onset non-singular. (c) phase direction: $g_{\theta\theta}$ finite ($2\text{--}3\times$ matched Gaussian).

We construct a testbed whose mean rotation rate is *analytically known*: the Stuart–Landau normal form $\dot{r} = \mu r - r^3$, $\dot{\theta} = \omega_0$, for which $r \rightarrow \sqrt{\mu}$ and θ rotates at exactly ω_0 . Over an observation window T_{obs} the winding number is therefore known exactly, $W = \omega_0 T_{\text{obs}} / 2\pi$, with $f_0 = \omega_0 / 2\pi$ placed inside the analysis band $[0.5, 5.0]$ Hz. The configuration is $N = 30$ dimensions and $T_{\text{obs}} = 200$ s ($M = 20,000$ samples), matching the production settings, and the screening statistic is the same band-integrated $\Im[S_{ij}]$ used in Phase 1.

a. *Recovery.* Across $f_0 \in \{0.6, 1.0, 2.0, 3.0, 4.5\}$ Hz and six seeds, the measured winding number recovers the injected value to a relative error $< 0.01\%$ in every realization, and the region-of-interest test fires in 30/30. The positive control therefore holds: the null findings reported in §IV A reflect the absence of collapse in the testbed, not an incapable estimator or protocol.

b. *Circulation versus delay.* We next compare three arms in which only the relation between the inspected pair differs: CIRC (a genuine rotation, phase offset constant at $+\pi/2$), DELAY (a passive lag, $y(t) = x(t - \Delta)$ with $\Delta = 0.35$ s, whose phase is *linear in f*), and INDEP (an independent oscillator). All three arms produce the same qualitative verdict: the first two fire (6/6 each, threshold ratios 33.2 and 26.7) while the third sits at the null (ratio 0.79). Sweeping the coherence, the I_{01} values attainable by the two firing arms overlap substantially, $[0.71, 26.80]$ against $[0.26, 26.80]$ for CIRC and DELAY respectively. Because the band integral $I_{ij} = \int \Im[S_{ij}] df$ is a *scalar*, it cannot encode whether the phase is constant in f or linear in f . A threshold-based binary decision is therefore *not* able to separate genuine circulation from a passive lag; this is the first quantitative justification for making Phase 2 (intervention) mandatory rather than optional.

c. *Power.* Detection was tracked while degrading phase coherence via phase diffusion D_θ on θ . The detection rate falls $1.00 \rightarrow 0.00$ and the threshold ratio $33.2 \rightarrow 0.77$ as D_θ is increased from 0 to 100, while the negative control stays at its nominal rate throughout. Notably it is *phase* diffusion, not amplitude noise, that erodes detectability: amplitude noise leaves θ untouched and hence leaves I_{ij} nearly unchanged, so an amplitude-noise sweep yields a flat curve. The winding-number accuracy degrades in step with detectability (relative error $\sim 1.5\%$ at $D_\theta = 50$, rising to 25–55% at $D_\theta = 100$), confirming that accuracy and power are governed by the same phase coherence.

d. *Two methodological by-products.* First, the production setting $\text{nperseg} = 256$ gives $\Delta f = 0.391$ Hz, and the $1/f$ floor then dominates the band: the spectral argmax always snaps to the lowest in-band bin. Single frequency recovery is therefore *not* available at the production setting (with $\text{nperseg} = 2048$ the injected f_0 is recovered to within 0.6%), which is consistent with the protocol’s use of the band integral alone and the absence of any peak-frequency criterion. Second, for narrowband channels the usual iid white-noise null is anti-conservative: it misclassifies 6/6 negative controls, whereas a *matched* null—the same marginal statistics without phase coupling—brings the false-positive rate back to its nominal level. Narrowband testbeds should calibrate against a matched null.

V. DISCUSSION

A. The metric-degeneracy prediction is refuted under identifiable conditions

Across **all tested boundary axes** ($\gamma, c_s, D \rightarrow 0, c_{\text{mem}}$) $\times$ **all coordinate systems** (Cartesian pairs, PCA $d = 2\text{--}5$, polar (r, θ)) $\times$ **the calibrated KDE** (and, qualitatively only, a k -NN cross-check, which is excluded from the quantitative comparison for the reason given in Sec. IV E), no information-geometric singularity ($\lambda_{\min} \rightarrow 0 \wedge \det g \rightarrow$

$0 \wedge \kappa \rightarrow \infty$) appears. Wherever evaluated, 95% CIs exclude zero—including the global minimum point $\gamma = 0.05$, $d = 5$ with CI [0.027, 0.032].

Why the prediction fails: a derived replacement, not a post-hoc excuse. The degeneracy condition is a statement about the **effective force**, not about the symmetry of the density. For a stationary Fokker–Planck system, $\nabla \cdot \mathbf{J} = 0$ with $\mathbf{J} = \mathbf{b}\rho - D\nabla\rho$ gives $\nabla \ln \rho = \mathbf{F}/D$ where $\mathbf{F} \equiv \mathbf{b} - \mathbf{J}/\rho$. With $g_{ij} = \langle \partial_i \ln \rho \partial_j \ln \rho \rangle = -\langle \partial_i \partial_j \ln \rho \rangle$,

$$\lambda_{\min} = 0 \iff v \cdot \mathbf{F} \equiv 0 \text{ along the trajectory,} \quad (7)$$

for some unit vector v . *Degeneracy is the null-direction condition on $\mathbf{F}$* , and a rotation is a coordinate transformation that drops out of $\mathbf{F}$; the only term that can supply a tangential component is a torque that does *not* co-rotate with the circulation. The drive of this system is a **lab-frame-fixed** antisymmetric coupling, so it does not co-rotate, $\lambda_{\min} > 0$ is preserved, and the prediction fails — but for a structural reason that can be stated in advance, not because a symmetry was “over-assumed.”

Two further corrections follow from the same derivation. (i) The claimed scaling of $\det g$ is **not coordinate invariant**: under $x = x(q)$, $\det g_{ab} = (\det J)^2 \det g$, so a “universal exponent” for $\det g$ is unverifiable without naming the variable it scales with; $\lambda_{\min} = 0$ (a null eigenvalue) *is* invariant, and $\lambda_{\min} \propto \xi_{\text{soft}}^{-2}$ follows from a scaling ansatz with an exponent fixed at -2 ($\det g \propto \xi^{-2d}$). (ii) In the phase direction, a first-order expansion about a uniform rotation ω_0 gives $g_{\theta\theta} = \omega_1^2 / [2(D_\theta^2 + \omega_0^2)]$ rather than $\omega_1^2 / (2D_\theta^2)$: **a uniform rotation screens angular anisotropy**. A test system with coherent rotation confirms this form (measured/predicted 0.894–1.118 over the resolvable range, ω_0 exponent -1.294 measured vs. -1.225 predicted) and rejects the ω_0 -independent form by a factor 2.7×10^4 over a 2.5-decade sweep. **Scope caveat**: that test system possesses a single coherent ω_0 , which the present spatiotemporal system does not, so only the *structure* (co-rotating drive $\Rightarrow$ degeneracy; fixed drive $\Rightarrow$ non-degeneracy) is transferable, not the quantitative scaling. The test-system verification code, the negative-control run that preceded it, and their complete logs are listed in the Data and Code availability statement below.

B. A coordinate-dependent criterion: why $\det g$ cannot carry a universal exponent

One structural defect in the published criterion is worth separating from the empirical result, because it would have invalidated the comparison *independently of any numerical evidence*.

The Fisher information metric is a rank-2 covariant tensor, so under $x = x(q)$ it transforms as $g_{ab} = J^T g J$ with $J = \partial x / \partial q$, and consequently

$$\det g_{ab} = (\det J)^2 \det g. \quad (8)$$

The determinant is therefore **not a coordinate invariant**: it picks up an arbitrary Jacobian factor. A statement of the form “ $\det g \rightarrow 0$ with a universal exponent” is thus not merely hard to test, it is ill posed until the variable with respect to which the scaling is taken has been named — and different choices of coordinates are related by J , so two legitimate coordinate systems can differ in their $\det g$ by an arbitrary factor. A quantity whose value can be changed by a relabelling of coordinates cannot carry a physical critical exponent.

The coordinate-invariant statement is the existence of a null eigenvalue, $\lambda_{\min} = 0$, and this is the criterion adopted above. Under a scaling ansatz $x \rightarrow \xi x$ with g the Fisher metric of the rescaled problem, $g \rightarrow \xi^{-2} g$ in the natural (unit-noise) units, so the soft eigenvalue obeys $\lambda_{\min} \propto \xi_{\text{soft}}^{-2}$ with the exponent fixed at -2 per direction, and $\det g \propto \xi^{-2d}$ only as a derived statement about a specified coordinate volume. The exponent -2 is thus a property of the scaling form, not a fitted universal index; the soft length $\xi_{\text{soft}} \sim \lambda_{\min}^{-1/2}$ is the physically meaningful object, and at the ROI working point ($\lambda_{\min} = 0.207$) it is $\xi_{\text{soft}} \simeq 2.2$, i.e. of order one, so the criterion cannot be near its singular point there at all.

Two practical consequences are already visible in our own data. First, the polar metric $\text{diag}(g_{rr}, 0)$ has rank one for *every* distribution by construction, and under the Cartesian transformation $r = \sqrt{x^2 + y^2}$, $\theta = \arctan(y/x)$ it stays rank one, so $\lambda_{\min} = 0$ in polar coordinates is a tautology and carries no information about the state of the system; the nonzero $\lambda_{\min} = 4$ we do measure there is the KDE noise floor, not a physical degeneracy. Second, reporting $\det g$ without the coordinates makes cross-paper comparison meaningless, which is why this work reports $\lambda_{\min}$ and κ as the primary invariants and treats $\det g$ as diagnostic only.

C. The operative discriminator: density sharpness

- The **matched-Gaussian ratio** of $\lambda_{\min}$ separates the ROI (≈ 1 ; Gaussian-flat) from EQ/DD ($\gg 1$; needle densities) by 6–9 orders of magnitude on a $\lambda_{\min}$ basis (KDE: 7–9, ROI 0.207 vs EQ 2.51×10^6 vs DD 1.30×10^8 ;

k-NN: 6–7, ROI 2.31 vs EQ 3.36×10^6 vs DD 5.97×10^7 ; the det-based gap is 14–17), independent of bandwidth and dimension, and—as a *qualitative* statement only—of density estimator; the k -NN cross-check is excluded from the quantitative comparison because its resolution floor overestimates $\lambda_{\min}$ by $24\text{--}231\times$ in the ROI regime (Sec. IV E); the manifold-dimension sweep ($d = 2\text{--}5$) and the alternative boundary axes leave this ordering unchanged.

- Physical interpretation: the non-equilibrium drive flattens and delocalizes the stationary density (circulation diffuses density through phase space, eroding pinned structures); equilibrium/overdamped systems concentrate density on small sets.
- **Recommended NEIP indicators:** $\lambda_{\min}(\text{emp})/\lambda_{\min}(\text{matched-G})$ or $\log_{10} \lambda_{\min}$, combined with intervention responsiveness (§IV B).

D. Methodological lessons

1. **det g is meaningless without dimension normalization:** it decays exponentially in d ($\det g = \prod \lambda_i$); compare $\lambda_{\min}$, $\det^{1/d}$, or matched-G ratios across dimensions.
2. **Estimator bias is systematic and regime dependent:** k -NN overestimates $\lambda_{\min}$ by $24.4\times$ at our ROI working point ($\kappa = 4.83$, true $\lambda_{\min} = 0.207$), rising monotonically to $231\times$ as $\lambda_{\min} \rightarrow 10^{-3}$, because the bias is a resolution floor rather than a calibration offset; the calibrated KDE returns 0.195 ($0.942\times$) at the same point. Absolute thresholds must be calibrated per estimator against a matched-Gaussian null, and an estimator whose bias exceeds the quantity being measured must be dropped from the quantitative comparison entirely.
3. **Validate the continuous limit first:** the legacy “signal” arose from a sign inversion plus a missing Δt . Report the three checks— Δt convergence, clip statistics, stationarity (with burn-in truncation)—alongside any Phase-1 verdict.
4. **Bootstrap and record-length scaling bound finite-sample spurious singularities:** both are standard equipment here. Over the record length $M \in \{3000, \dots, 20000\}$ (written M because N denotes the spin count throughout), $\lambda_{\min}$ stays strictly positive and saturates *non-monotonically*: $0.1691, 0.1772, 0.1734, 0.1850, 0.1974, 0.2036$ —a total change of $+20\%$, not a divergence and not a decay toward zero. The point of the check is that $\lambda_{\min}$ neither escapes upward without bound nor approaches zero, so the non-degeneracy conclusion is not an artifact of the record length.

E. Prospects and limits

The protocol is strongest under quasi-stationarity, full state observation, and intervention feasibility. Under partial observation, strong non-stationarity, or intervention impossibility, the causal strength of Phase 2 degrades and additional machinery (latent-space embedding, time-varying spectral estimators, passive causal discovery) is required. Short series / coarse sampling degrade both Welch resolution and KDE accuracy (the “forbidden operating regime”). The full $d = 30$ configuration-space metric remains unevaluated (curse of dimensionality); this protocol covers submanifolds up to $d \leq 5$. The sharpness transition (jump of $\lambda_{\min}$ to $10^5\text{--}10^8$ around $\gamma \approx 0.5\text{--}0.75$) awaits systematic characterization (scaling laws, finite-size effects).

VI. CONCLUSIONS

1. The corrected SPDE (continuous-time discretization) is clip-free, Δt -convergent, and stationary, providing a sound testbed for NEIP verification.
2. Spectral-imaginary screening identifies the ROI at $\gamma \lesssim 0.2$ seed-independently; **causality is established** by intervention ($p = 0.0024$) and negative control ($p = 0.47$).
3. The **metric-degeneracy prediction of Paper III is refuted** [7] across the full falsifiable range (axes, coordinates, dimension, estimator); the theory must re-examine its implicit symmetry assumptions.
4. **Density sharpness is the robust NEIP discriminator:** the non-equilibrium drive flattens the density; equilibrium/overdamped regimes needle it.

5. The complete protocol (screening $\rightarrow$ intervention $\rightarrow$ geometry) delivers an implementable resolution of the three bottlenecks (identifiability, spurious singularities, computational explosion).
6. A **positive control** on a testbed with an analytically known winding number confirms that the screening stage recovers known circulation (relative error $< 0.01\%$) and retains power as coherence degrades; the null findings above therefore reflect the absence of collapse, not an inert estimator. The same control shows that the scalar band integral cannot separate a genuine circulation from a passive lag, which is why Phase 2 is required rather than optional.

VII. DATA, CODE, AND REPRODUCIBILITY

All numerics used the corrected scripts in the companion repository, with fixed seeds and deterministic ordering. The simulation and metric-estimation routines are `simulate_corrected.py` (SPDE) and `compute_geometry.py` (KDE–Fisher); the phase-1 screening sweep, phase-2 intervention sweeps, and phase-3 cross-checks are in `experiment1.fixed.py` through `experiment8.fixed.axes.phase.py`.

Result tables: `boundary_sweep_result.csv`, `highdim_metric_result.csv`, `axis_phase_result.csv` (including the KDE-matched Gaussian column `gauss_est`). Figures regenerated by `figure_maker.py`; data consistency verified by `figure_data_check.py` (ALL PASS, 2026-09-16).

Detailed per-experiment records are maintained in the companion research notes of this study; the working draft (bilingual, research notes) mirrors this manuscript.

The theory-verification experiments reported in Sec. V are released together with their complete logs: the re-designed run (Exp. 16; 276 s wall clock, 53 result rows), its post-analysis script, and the preceding run (Exp. 15) that *failed to reach a verdict* because the estimator noise floor exceeded the signal. Both are included deliberately: the negative result is the quantitative justification for the estimator redesign described in Sec. III C, and omitting it would present the successful run as though it had been designed correctly from the start. The driver scripts are shipped as `16_theory_verification.py` and `15_theory_verification.py`, with the result table `theory_verification16_result.csv` and the full stdout logs of both runs.

Significance Statement: *A previously published criterion for detecting nonequilibrium phase boundaries—a simultaneous vanishing of the Fisher-metric determinant, its smallest eigenvalue, and the divergence of the condition number—does not fire when tested against a controlled many-body testbed, even when the criterion’s own theory is presumed satisfied. Had this criterion been adopted uncritically, a broad region of parameter space would have been mislabeled as a phase boundary on the strength of finite-sample numerical artifacts. This work therefore replaces the metric-degeneracy signature with an operational one—density sharpness—that does separate the regimes, and shows that the protocol which screens for circulation is statistically sound once the multiplicity, bootstrap, and estimator-bias problems inherent in the original formulation are corrected. A positive control on a testbed with a known winding number confirms that the screening stage does recover circulation when it is present, so the null results reported here are a property of the systems studied rather than of the method. The failure is not, however, confined to the numerical side of the original criterion. The degeneracy condition itself is a statement about the symmetry of the stationary density, whereas the Fisher information metric of a Fokker–Planck system with $\nabla \cdot \mathbf{J} = 0$ and $\mathbf{J} = \mathbf{b}\rho - D\nabla\rho$ obeys $g_{ij} = -\langle \partial_i \partial_j \ln \rho \rangle$ with $\nabla \ln \rho = \mathbf{F}/D$ for the effective force $\mathbf{F} \equiv \mathbf{b} - \mathbf{J}/\rho$. A null eigenvalue of g requires $\mathbf{F}$ to vanish along a direction, so degeneracy occurs precisely when the nonequilibrium drive co-rotates with the circulation, and any lab-frame-fixed anisotropy—including the antisymmetric coupling used here—prevents it. The system studied here has a lab-frame-fixed drive, so $\lambda_{\min} > 0$ is preserved and the prediction fails for a structural reason that can be stated in advance. We further show that the commonly quoted universal exponent for $\det g$ is ill posed, because $\det g$ is not a coordinate invariant: under $x = x(q)$ it transforms as $\det g_{ab} = (\det J)^2 \det g$. Only the null-eigenvalue statement is invariant, and that is the criterion we adopt. The correction is quantitative and was verified on a separate controlled system with a uniform rotation rate, where the predicted rotational screening of the anisotropy agrees with measurement over more than two decades; we emphasize that the result transfers to the present many-body system as a condition, not as a scaling law, and we have not measured the corresponding component of the metric here. Because the diagnostic quantities (Fisher metrics from kernel density estimates, quadrature cross-spectra) are standard tools in time-series analysis of driven and active matter, both the corrected screening recipe and the revised form of the criterion transfer directly to other nonequilibrium phase-detection studies.*

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